Programming Fundamentals
Quiz-12

Problem 1 Matrix Sort

The matrix sort problem entails sorting based on a column in ascending or descending fashion while ensuring the order of the elements with respect to other rows, respectively.

- **void swapRows(int M[ ][COLS], int sr, int dr, int Cols)**

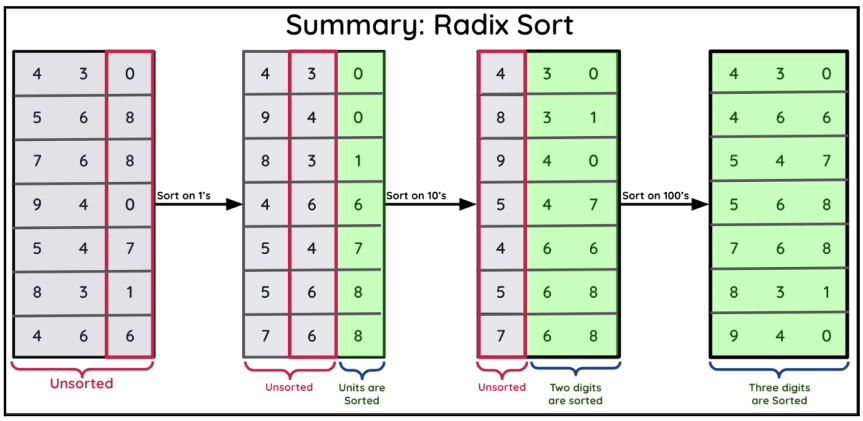
  Sample program

Matrix: 4 3 2 1 5 5 2 5 4 6 1 2 3 5 7 8 9 6 3 2 1 3 5 7 9	**Input:** Enter column index to sort: 3 **Output:** 4 3 2 **1** 5 8 9 6 **3** 2 5 2 5 **4** 6 1 2 3 **5** 7 1 3 5 **7** 9

Problem 2 Radix Sort

Consider an m x n matrix where each row constitutes a complete number. For example, take a look at the 3x 5 matrix below:

Sample Input:	Sample Output:
4 3 0 7 8 6 9 4 0 5 4 7 4 6 6	4 3 0 4 6 6 5 4 7 7 8 6 9 4 0



Details: you need to implement two functions:

- **void swapRows(int M[][maxCols], int cols, int ri, int rj)**
- **void bubbleSortBasedOnColIndex(int M[][maxCols], int rows, int cols, int ci)**

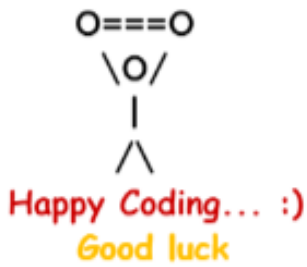
Problem 3 Set Matrix Zeroes

Given an m x n integer matrix matrix, if an element is 0, set its entire row and column to 0's.You must do it in-place.

Example:

1	1	1		1	0	1
1	0	1	→	0	0	0
1	1	1		1	0	1

Input: matrix = [[1,1,1],[1,0,1],[1,1,1]]
Output: [[1,0,1],[0,0,0],[1,0,1]]

0	1	2	0		0	0	0	0
3	4	5	2	→	0	4	5	0
1	3	1	5		0	3	1	0

Input: matrix = [[0,1,2,0],[3,4,5,2],[1,3,1,5]]
Output: [[0,0,0,0],[0,4,5,0],[0,3,1,0]]